Physiological response prediction based on Uncertainty-Awared Attention

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Introduction

Physiological responses are human's response to the environment, including heart rate, breath rate, body temperature etc. There are some connection between physiological responses. With the help of neural network and other tools, we can reveal the connection between them and make predictions based on them. Physiological response prediction is the technology to predict unkown physiological response with data already know. To make the prediction more reliable, prediction uncertainty is introduced.

Object

This study is aimed to predict predict breath rate with human body movement data and former breath rate.
To make the prediction more accurate, we want to add uncertainty-awared attention to the model.

Background

- **Attention Mechanism**

Attention mechanism is a method to assign weight. It is effective in both focusing the deep learning models on relevant features and interpreting them.

- **LSTM(Long Short Term Memory)**

It is a kind of RNN, added with cell state variable to help the network to deal with long sequences.

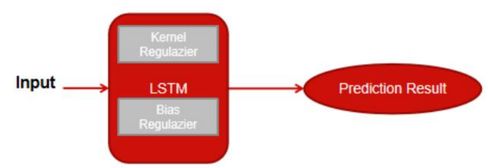
- **Bayesian Neural Network**

It is a combination of Bayesian statistics and deep learning. It can model uncertainty in the model. With the uncertainty, the model can output the prediciton reliability.

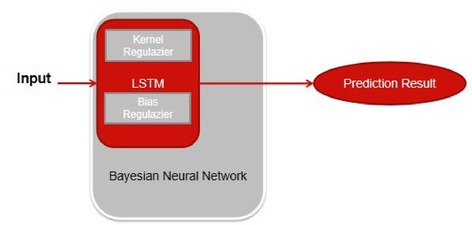
Models

We present three models: baseline 1, baseline 2, UAP

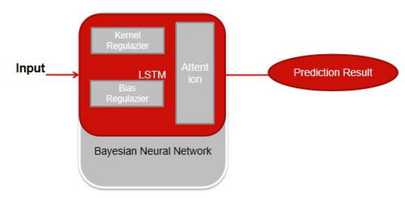
Baseline 1: LSTM model



Baseline 2: LSTM+BNN model



UAP(Uncertainty-Awared Attention Predictor)



Experiment

We used Fabian dataset for experiment:

- The dataset contains physiological responses and environment data, including breath rate, movement, body temperature.
- Data is collected from a healthy, 35-40 year old male subject over a period of four days and 7.0-8.0 hours per day.
- RMSE(Root Mean Squared Error) is used to measure how close it the prediction to real value, the lower the better

Result

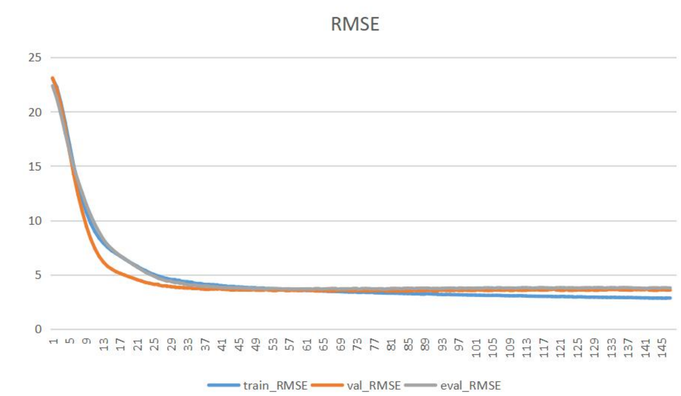


Figure 1: result of UAP

	LSTM	LSTM+BNN	UAP
RMSE	2.89471415	2.485112479	3.768299737

Table 1: result of three models

- Compare two baseline, we can find uncertainty can help improve prediction reliability
- The RMSE of UAP is higher than baseline models

Conclusion

The UAP model does not better than the baseline, two ways of explanations are presented for that:

- It is not very accurate to measure the attention uncertainty. So reliable prediction may be regarded as unreliable, so it will be abandoned. Some unreliable prediction may be regarded as prediction, so it will remain and decrease the RMSE eventually.
- The model hasn't be tuned good enough to outperform the baseline. Hyper parameters, like lambda in loss function, dropout prob.

Acknowledge

The work is completed in ARoS laboratory in North Calorina State University. The dataset set is collected and processed by people in ARoS laboratory. Special thanks to Sabrina for help with the work.